# Module 8: Trees

## Part 2: Bagging and Random Forest

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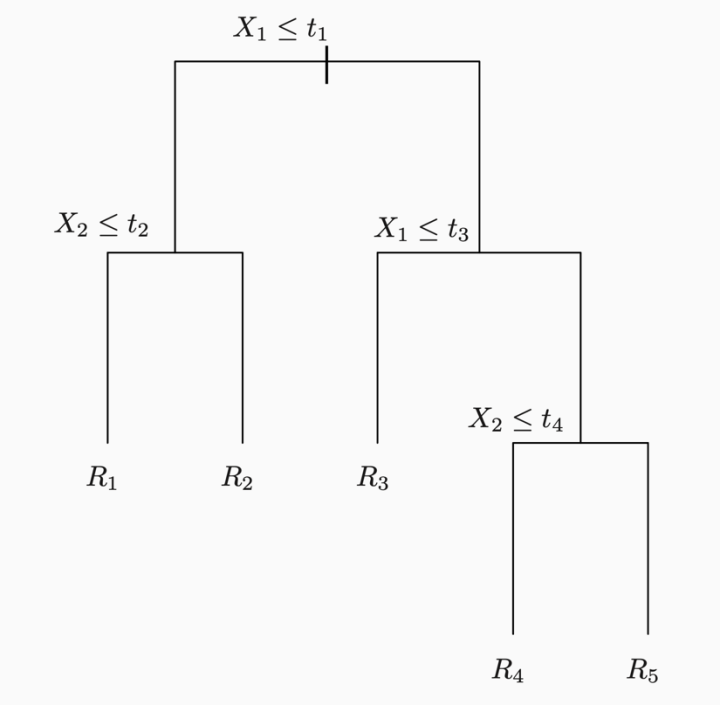
# Copyright and Attribution

Some lecture slides and instructional materials in this course are adapted from the following sources:

•*An Introduction to Statistical Learning: With Applications in R (Second Edition)*
Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani
Springer, 2021

•Online course materials developed by Trevor Hastie, Robert Tibshirani, and collaborators.

- Decision trees have **high variance**
  → small data changes → very different trees
- Goal: **reduce variance**




How can we reduce the variance ?

# Stats Review

Suppose we are given a set of $n$ **independent** variables $Z_1, \ldots, Z_n$. Each has their own variance $\sigma^2$.

- Variance of $Z_i = \sigma^2$
- Variance of $\bar{Z} = \sigma^2/n$

Averaging a set of observations reduces variance!

# Ensemble Methods

- An **ensemble method** is an approach to **combines many simple models** in order to obtain a single and potentially very powerful model.
- These simple models are sometimes known as **weak learners**, since they may lead to mediocre results on their own.
- **Tree-based ensembles:**
  - Bagging
  - Random Forests
  - Boosting

# Apply this logic to trees

- How to reduce variance in trees?
- Take the average of them!
- Idea: take **many** training sets from the **population**, construct a tree using each training set, and **average** the resulting predictions:

$$\hat{f}_{\text{bag}(x)} = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{b}(x)$$

Obvious problem with this approach?

# Bootstrap

- Generate $B$ different bootstrapped training sets.
- For the $b$th bootstrapped training set, we get a bootstrap sample and its predictions $\hat{f}^{(b)}(x)$.
- Average all the predictions to obtain:

$$\hat{f}_{\text{bag}(x)} = \frac{1}{B}\sum_{b=1}^{B} \hat{f}^{b}(x)$$

- This is called bagging.

Bootstrap: Sampling with replacement

# Bagging

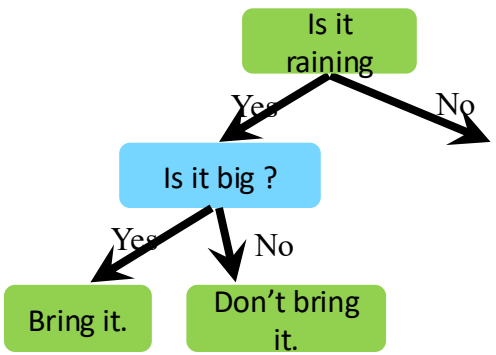
To apply bagging to regression trees:

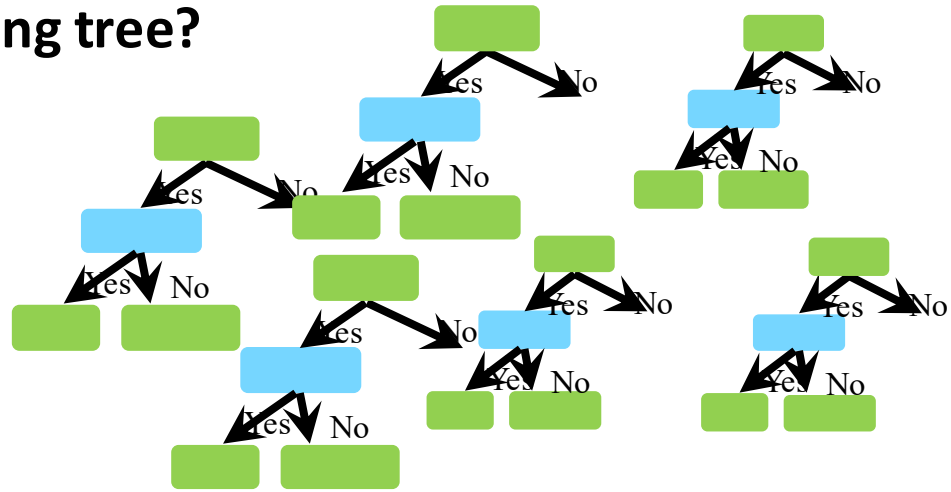
- Construct **B regression trees** using **$B$ bootstrapped training sets**, and average the resulting predictions.
- Each individual tree grown deep and **not pruned**.
- Each tree has high variance, but low bias.

- **Averaging these $B$ trees reduces the variance.**
- Improvements in accuracy by combining together hundreds or even thousands of trees into a single procedure.
- Number of $B$ is not critical with bagging. Use a value of B sufficiently large that the error has settled down.

# Bagging

Bagging improves prediction accuracy at the expense of interpretability.
**How do we interpret the resulting tree?**



- Overall summary of the **importance** of each prediction using the RSS (regression trees) or Gini index (classification trees).
- **Regression trees**: record the total amount that the **RSS** is decreased due to splits over a given predictor, averaged over all $B$ trees. A large value indicates an important predictor.
- **Classification trees**: record the total amount the **Gini index** is decreased by splits over a given predictor, averaged over all $B$ trees.

# Bagging

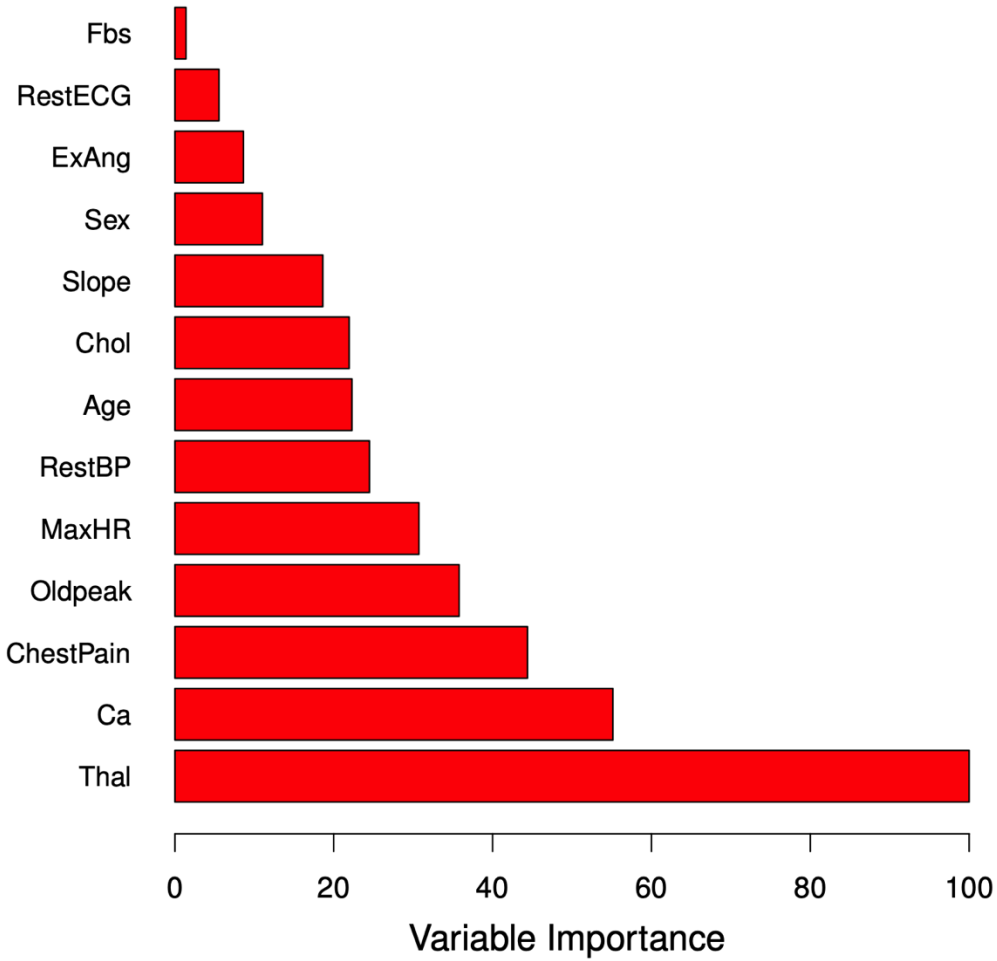



Fig 1. A variable importance plot for the Heart data.

# Bagging

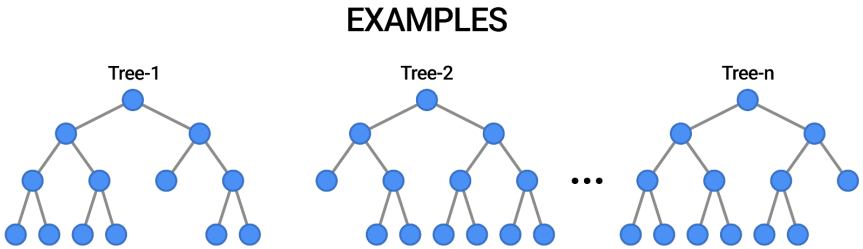
Problem with this approach?

# Random forest

Random forests provide an improvement over bagged trees by decorrelating trees.

- Each time a split in a tree is considered, a **random sample of $m$ predictors** is chosen as candidates from the full set of $p$ predictors.
- Split is only allowed to use those $m$ predictors.
- A new sample of $m$ predictors is taken at each split.
- Typically we choose $m \approx \sqrt{p}$.
- This process of decorrelating the trees has the effect of reducing the variance.

- Bagging: many similar trees
- Random forest: many more **diverse** trees



# Implementation

See R script: random_forest.R